



Lab 06

Impedance, Transient, and Steady State Response of Series RLC Circuits

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6.1 Objective

The objectives of this lab are:

- To find the impedance of series RLC circuit and analyze its relationship with frequency.
- To observe time domain response of RLC circuit to step input.

6.2 Pre-Lab Task

Task 1:

Q1. For a series and parallel RLC circuit what are the conditions for overdamped, under-damped and critically damped circuits, draw the waveform for all three conditions with step voltage as an input function. (5 points)

Q2. Calculate the theoretical values of $X_C \angle \phi$, $X_L \angle \phi$ and $Z \angle \phi$ for input frequencies provided in Table 6.2. Provide all the calculation on a separate A4 sheet. (10 points)

6.3 Introduction

The series RLC circuit has a single loop with the instantaneous current flowing through the loop being the same for each circuit element. Since the inductive and capacitive reactance, X_L and X_C are functions of the supply frequency, the sinusoidal response of a series RLC circuit will therefore vary with frequency f . The instantaneous voltage across a pure resistor, V_R is “in-phase” with the current. The instantaneous voltage across a pure inductor, V_L ‘leads’ the current by $+90^\circ$. The instantaneous voltage across a pure capacitor, V_C ‘lags’ the current by -90° . Therefore, V_L and V_C are 180° ‘out-of-phase’ and in opposition to each other.

This circuit configuration is often used in various applications, including signal filtering, impedance matching, and resonant circuits.

6.4 Series RLC Circuits

A series RLC circuit exhibits different behaviors at different frequencies due to the combined effects of the resistor, inductor, and capacitor. The resonance phenomenon is an essential feature of these circuits and is particularly useful in applications like tuning, filtering, and frequency response shaping.

Behavior of the Series RLC Circuit at Different Frequencies:

- 1) **DC (Direct Current):** When a direct current is applied (frequency = 0), both the inductor and capacitor act as short circuits (ideal inductor) and open circuits (ideal capacitor), respectively.
- 2) **Low Frequencies:** At lower frequencies, the inductive reactance (X_L) is lower than the capacitive reactance (X_C), resulting in a net capacitive impedance.
- 3) **Resonant Frequency:** At a specific frequency known as the resonant frequency (f_{res}), the inductive reactance equals the capacitive reactance, and they cancel each other out. At this frequency, the circuit is purely resistive, and the impedance (total opposition to current) is minimized. The resonant frequency is given by $f_{res} = 1 / (2\pi\sqrt{LC})$.
- 4) **High Frequencies:** At higher frequencies, the capacitive reactance decreases and the inductive reactance increases, resulting in a net inductive impedance.

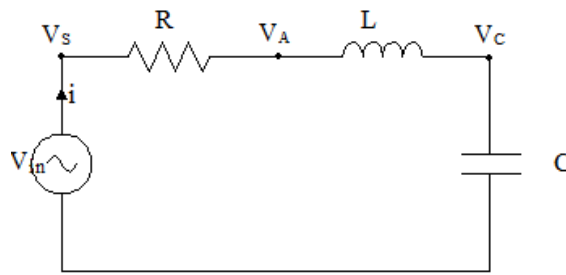


Figure 6.1: Series RLC Circuit

Formulae

$$X_c = \frac{-j}{2\pi fC} \quad (1)$$

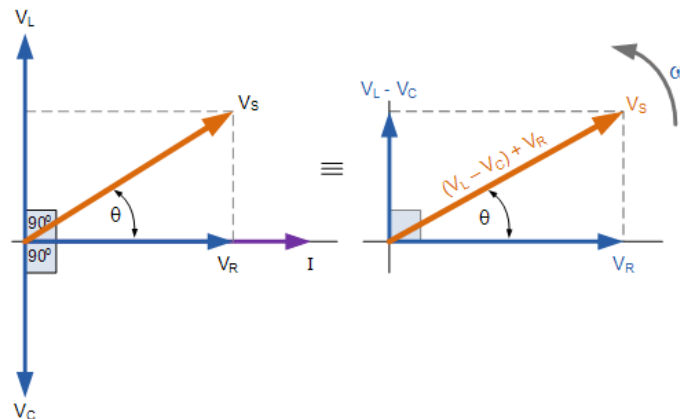
$$X_L = j2\pi fL \quad (2)$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2} \quad (3)$$

$$\phi = \left(\frac{X_L - X_C}{R} \right) \quad (4)$$

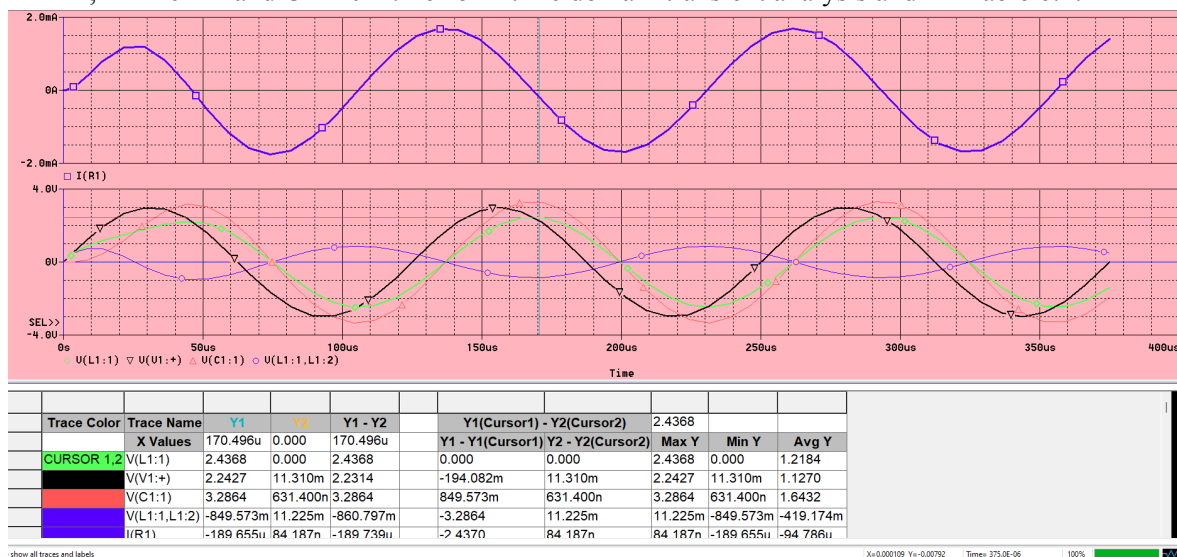
$$\phi_{sim} = \Delta t \times 360 \times freq \quad (5)$$

Phasor Diagram



Task 2: To perform time domain transient analysis of series RLC Circuits on PSpice

1. Simulate the circuit as shown in Figure 6.1 on PSpice. Keep $V_{in} = 3$ (VSIN) at 8kHz, $R = 1$ k Ω , $L = 10$ mH and $C = 10$ nF. Perform time domain transient analysis and fill Table 6.1.



Note: Consider the second cycle of the waveform when noting down the measurements.

2. Calculate the deviation between simulated and theoretical values of X_C and X_L .

Table 6.1

I_C	1.6801	I_L	1.6801
V_C	3.3226	V_L	0.8495
Δt_C	12.064	Δt_L	12.064



Φ_C	34.744320	Φ_L	34.744320
X_C	V_C/I $= 3.3226/(0.0016977)$ $= 1957.12$	X_L	V_L/I $= 0.849573/0.0016977$ $= 500.42$
Deviation	32.28	Deviation	2.274

3. According to frequencies mentioned in Table 6.3 perform Transient Analysis and record $V_C \angle \phi$, $V_L \angle \phi$, $I_T \angle \phi$ and Δt of each parameter w.r.t to $V_{in} \angle \phi$. Make sure Δt is noted with correct sign (i.e., keeping V_{in} as reference) it will result in correct signs for ϕ . If the waveform leads the waveform of V_{in} , ϕ will be positive otherwise it will be negative. For capacitor voltage use the $V(C1:2, C1:1)$

Table 6.2

Freq (Hz)	X_C	X_L	$Z \angle \phi$
8 k	1989.4 $\angle$ -90	502.7 $\angle$ 90	1791.8 $\angle$ -56.07
12 k	1326.3 $\angle$ -90	753.98 $\angle$ 90	1152.2 $\angle$ -29.78
16 k	994.7 $\angle$ -90	1005.3 $\angle$ 90	1000.6 $\angle$ 0.607
20 k	795.78 $\angle$ -90	1256.6 $\angle$ 90	1101.1 $\angle$ 24.74
24 k	663.15 $\angle$ -90	1508 $\angle$ 90	1309.1 $\angle$ 39.2
30 k	530.5 $\angle$ -90	1885 $\angle$ 90	1683.6 $\angle$ 53.56

Table 6.3

Freq (Hz)	$V_{in} \angle \phi$	$V_C \angle \phi$	Δt_C	$V_L \angle \phi$	Δt_L	$I_T \angle \phi$	Δt	$X_C \angle \phi$ V_C/I_C	$X_L \angle \phi$ V_L/I_L	$Z \angle \phi$ V_{in}/I_T
8 k	2.9717	3.3226	12.064	0.8495	-12.064	0.0016977	19.368	1957.12	500.42	1750.427 $\angle$ -55.78
12 k	2.9533	3.4432	10.527	1.9903	-8.4293	0.0026275	6.622	1310.45	757.49	1124.000 $\angle$ -28.61
16 k	2.9876	2.9333	7.8997	3.0617	-7.2313	0.0029697	-0.468	987.74	1030.98	1006.0275 $\angle$ 2.696
20 k	2.9833	2.0634	5.7890	3.4471	-5.6782	0.0026297	-3.864	784.50	1310.83	1134.464 $\angle$ 27.82
24 k	2.8610	1.3999	3.4678	3.3994	-4.238	0.0021320	-5.078	656.61	1594.47	1341.932 $\angle$ 43.873

30 k	2.9302	0.868001	1.8966	3.050	-3.255	0.0017235	-4.982	527.5	1884.7	1683.60 $\angle$ 53.805
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Task 3: To perform AC sweep analysis of series RLC Circuits on PSpice

Perform AC sweep analysis on PSpice to identify the resonant frequency point as mentioned in Figure 6.2. Attach the graphs of following parameters along with brief comment on the trend and its reason.

1. Z (magnitude) vs Frequency
2. Z (Phase) vs Frequency
3. I_T vs Frequency
4. X_L , X_C vs Frequency

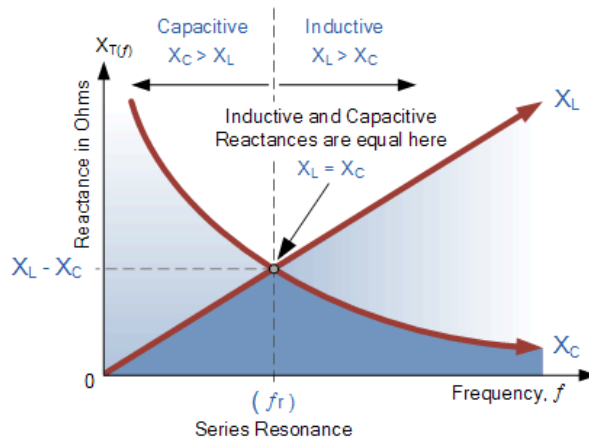
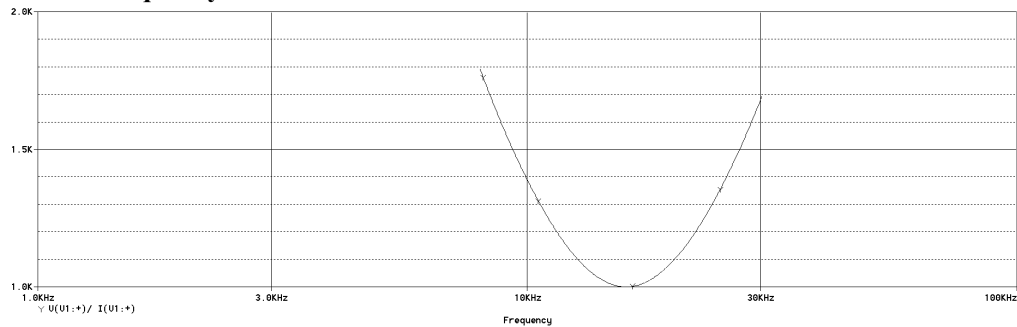


Figure 6.2: Series resonance frequency curve

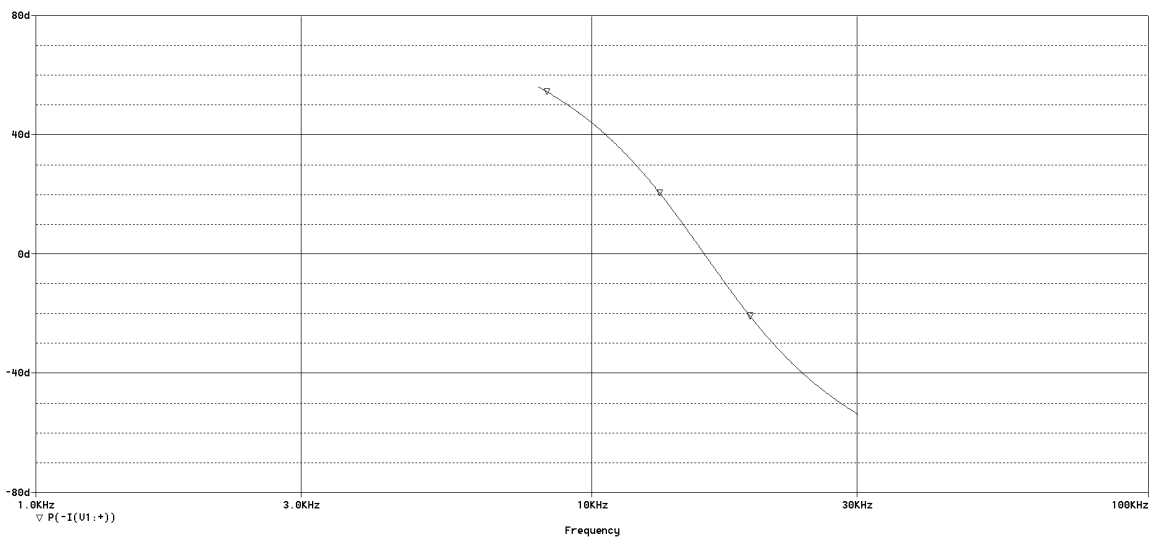
Z (Magnitude) vs Frequency:

resonant frequency: 15.947KHz



comments: As frequency moves away from resonant frequency (where $X_L = X_C$), impedance increases because either X_L (at higher frequencies) or X_C (at lower frequencies) dominates.

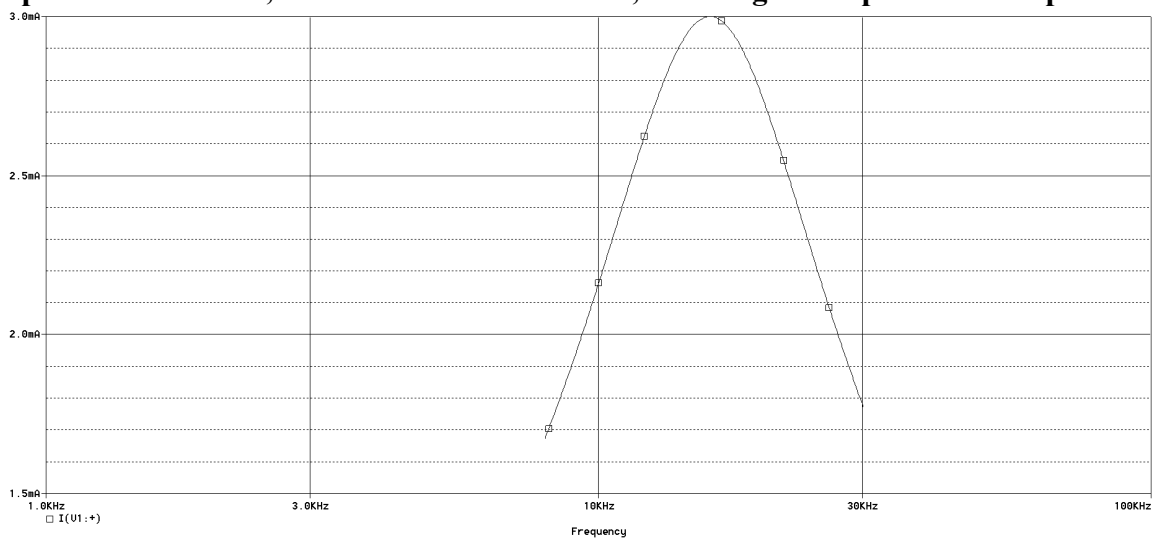
Z (Phase) vs Frequency



comments: Z (phase) is found using $-I_T$ because $Z = V_{in}/I_T$ and the phase of V_{in} is 0 so the impedance phase is the negative of I_T phase. At resonance, the phase is zero because the circuit is purely resistive.

I_T vs Frequency

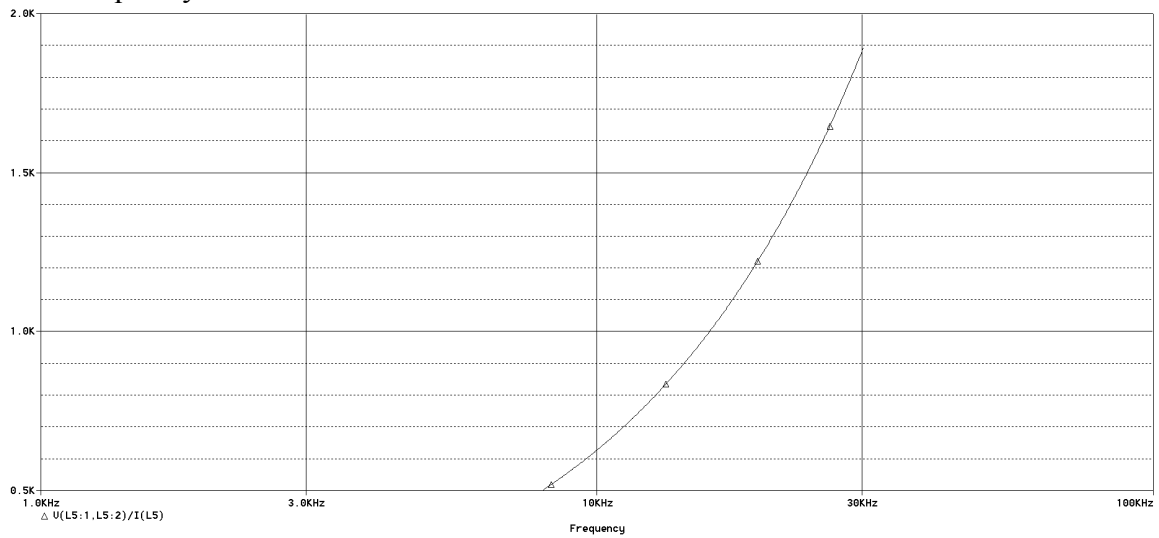
comments: The current in a series RLC circuit is inversely related to the impedance. when the impedance is at its minimum, the current is maximum. hence, this graph shows a peak at resonant freq (15.947 kHz). As you move away from this peak, the impedance increases, and the current decreases, resulting in the parabolic shape.



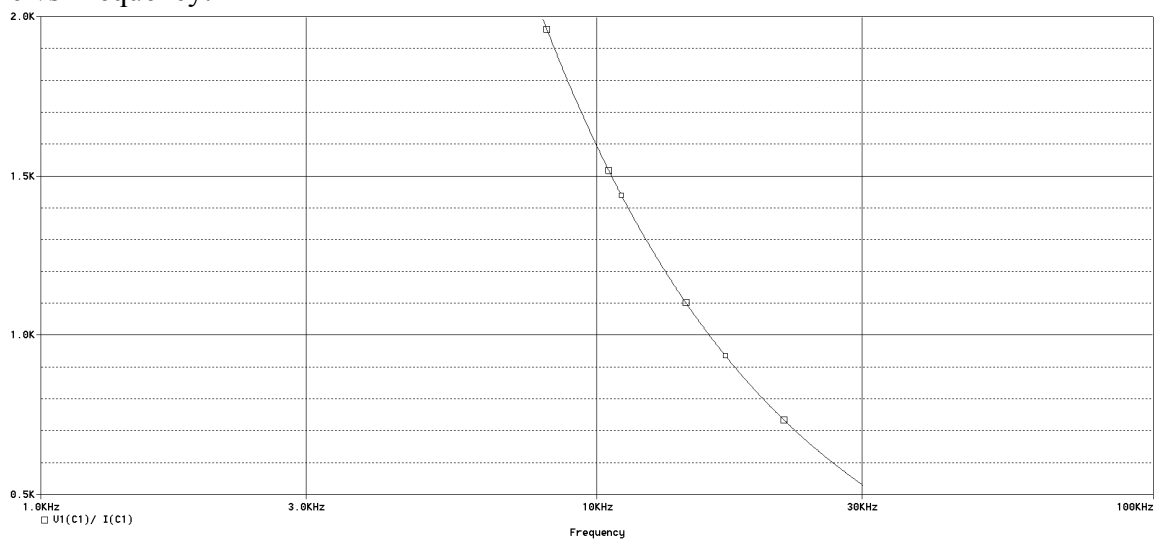
X_L , X_C vs Frequency



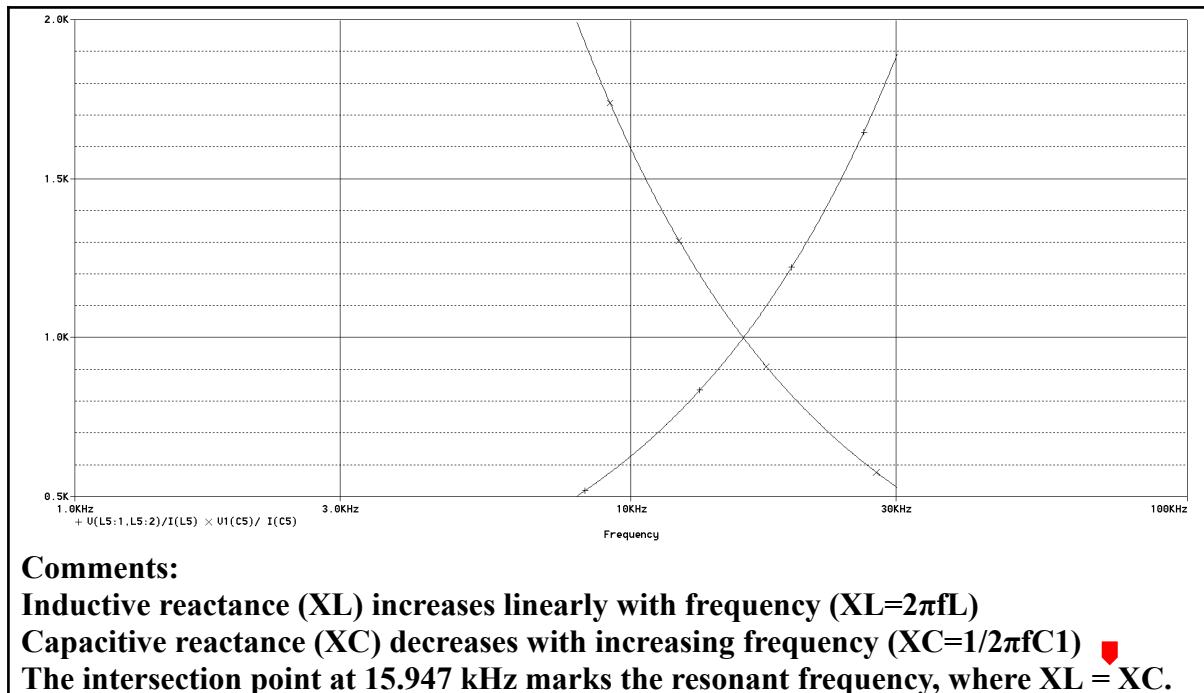
Xl vs Frequency:



Xc vs Frequency:



intersection point = 15.947kHz



6.5 Time Domain Response of RLC Circuits to Step Input

The series RLC circuit has a single loop with the instantaneous current flowing through the loop being the same for each circuit element. Since the inductive and capacitive reactance, X_L and X_C are functions of the supply frequency, the sinusoidal response of a series RLC circuit will therefore vary with frequency f . The instantaneous voltage across a pure resistor, V_R is “in-phase” with the current. The instantaneous voltage across a pure inductor, V_L ‘leads’ the current by $+90^\circ$. The instantaneous voltage across a pure capacitor, V_C ‘lags’ the current by -90° . Therefore, V_L and V_C are 180° ‘out-of-phase’ and in opposition to each other.

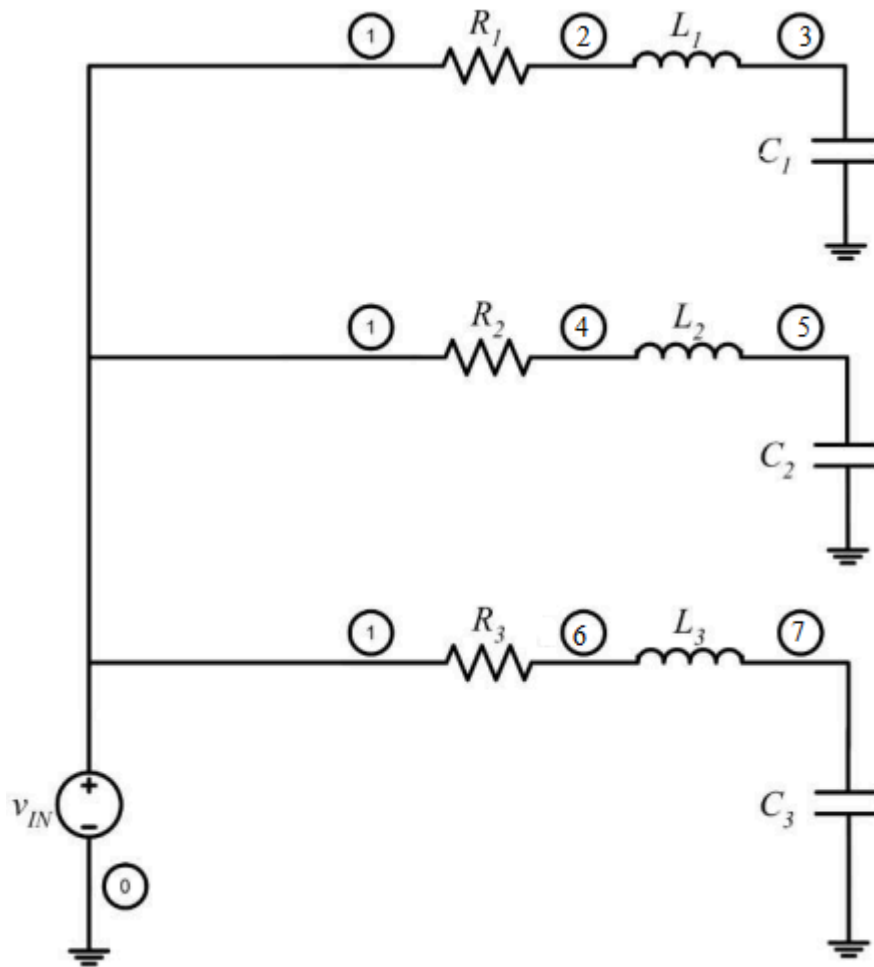


Figure 6.3

Formulae:

$$L \frac{di(t)}{dt} + Ri(t) + V = V_{IN}$$

Substituting $i = C \frac{dV}{dt}$ and rearranging terms

$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_{IN}}{LC}$$

Characteristic Equation is given by

$$s^2 + \frac{1}{RC} s + \frac{1}{LC}$$

Roots of the equation are given by



$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

where,

$\alpha = \frac{R}{2L}$	Damping Factor
$\omega_0 = \sqrt{\frac{1}{LC}}$	Resonant Frequency
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	Damping Frequency

The solution to above equation has two components: **the transient response and the steady-state response** that is,

$$v(t) = v_t(t) + v_{ss}(t)$$

The steady-state response is the final value of $v(t)$

$$v_{ss}(t) = v(\infty)$$

The transient response for the overdamped, underdamped, and critically damped cases are

$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$	(Overdamped)	$\alpha > \omega_o$
$v_t(t) = (A_1 + A_2) e^{-\alpha t}$	(Critically damped)	$\alpha = \omega_o$
$v_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$	(Underdamped)	$\alpha < \omega_o$

Task 4: To perform time domain analysis of RLC circuits to Step Input

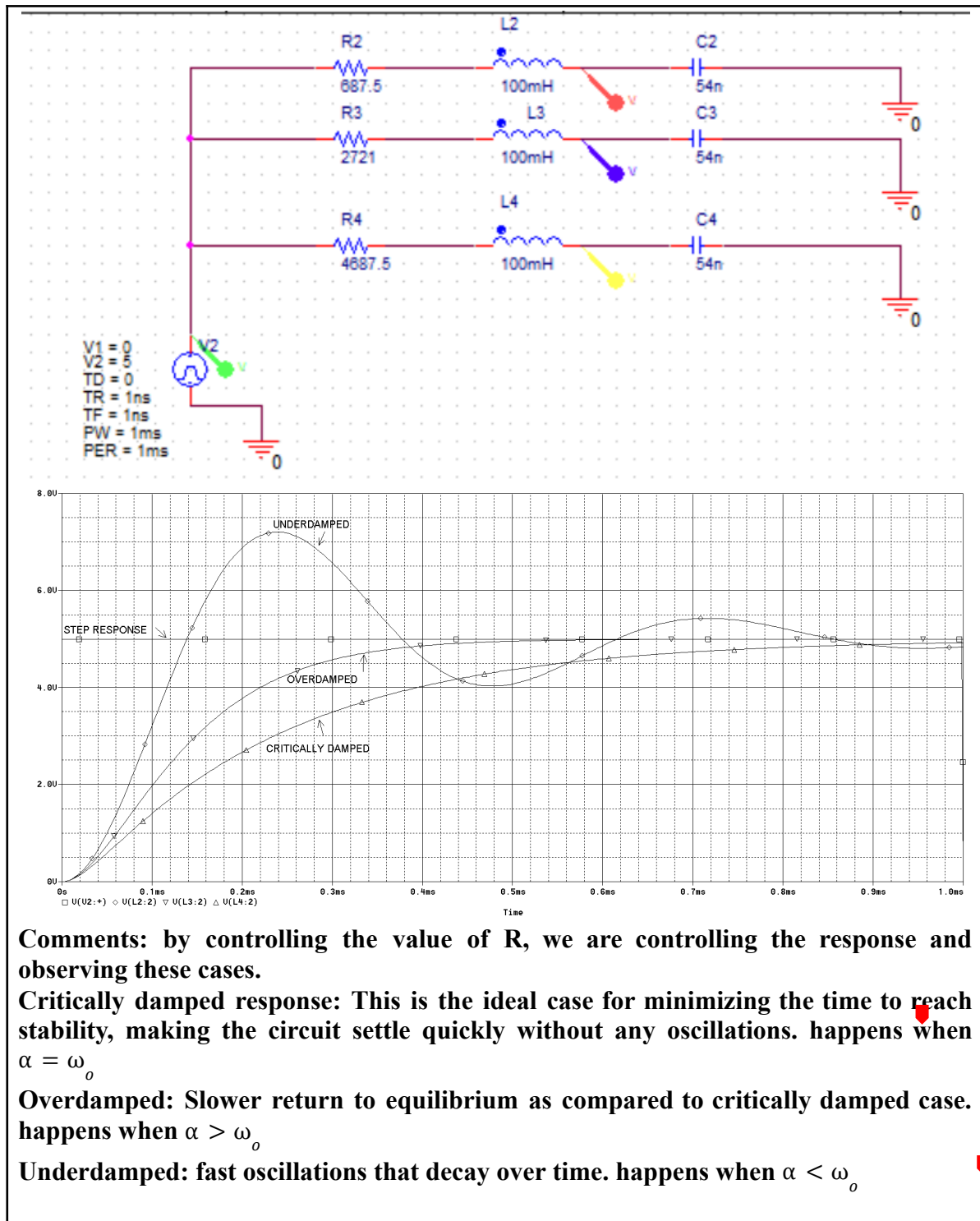
- Figure 6.3 shows three independent circuits, connected to a common input signal. This configuration will enable you to observe the under damped, critically damped and over damped cases simultaneously.
- Simulate the circuit shown in Figure 6.3, using the following values:
 - $V_{IN} = 1V$ step



- $L_1 = L_2 = L_3 = 100mH$
- $C_1 = C_2 = C_3 = 54nF$
- $R_1 = \frac{XXX+400}{2} \Omega$
- $R_2 = 2721\Omega$
- $R_3 = R_1 + 4000\Omega$

XXX represent the last three digits of your Student ID

3. For the step voltage source use the VPULSE component from the Source library. Specify appropriate values for the V1, V2 (you can take V1=0 and V2=5) and TD parameters of VPULSE (V1 and V2 correspond to the minimum and maximum voltages of the pulse, respectively, and TD corresponds to the delay from time zero to the first rising edge).
4. To specify the time range for the simulation, go to PSpice / New Simulation Profile and choose 'Time domain (transient)' analysis type. Make sure you specify an appropriate time range to display the outputs clearly. You may need to iterate through a number of times to obtain the required output plot (in that case, go to PSpice / Edit simulation profile to edit the settings). We need to see the transient response it can be observed in the start of the waveform $(1/10000)^{th}$ of the time period.
5. Simulate the circuit and, on the same plot, graph the input signal VIN, as well as the three output signals, V3, V5 and V7, as specified in Figure 6.3.
6. Identify which one of the circuits is under damped, which one is over damped, and which one is critically damped in the plot support your answers with equations of complete response of your circuit.
7. Attach the screenshot of PSPICE model and plot. Also, provide analysis on obtained plots below:





Frequently Asked Questions (FAQs)

Q1. Can the voltage drop across the inductor or the capacitor in a series LCR circuit be greater than the applied voltage of the AC source?

Ans: Yes; the voltage-drop across the inductor or the capacitor in a series LCR circuit can be greater than the applied voltage of the AC. source because as V_C and V_L have opposite faced, V_C or V_L maybe greater than V .



Assessment Rubric
Lab 06
Impedance, Transient, and Steady State Response of Series RLC Circuits

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Points Distribution

Task No.	LR 2 Simulation	LR5 Results (table) /Plots	LR6 Calculation	LR 10 Analysis
Task 1 (Pre-Lab)		/5	/10	
Task 2	/5	/20	/5	-
Task 3	/2.5	/10	-	/10
Task 4	/2.5	/10	/5	/5
Total	/10	/45	/20	/15
CLO Mapped	CLO 2	CLO 2	CLO 2	CLO 2

Affective Domain Rubric		Points	CLO Mappe d
AR 7	Report Submission	/ 10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR7	Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.

